LABSHEET-06

Q1. Create a directory named lab6 and a text file named permissions.txt inside it.

```
mkdir lab6
touch lab6/permissions.txt
```

Q2. Display the long listing of permissions.txt and identify the owner, group and current rwx permissions.

```
ls -l lab6/permissions.txt
```

Q3. Change the permissions of permissions.txt using symbolic notation so that the owner has read, write and execute permissions, the group has read and write permissions, and others have no permissions.

```
chmod u=rwx,g=rw,o= lab6/permissions.txt
```

Q4. Change the permissions of permissions.txt using numeric mode 640 and verify the result.

```
chmod 640 lab6/permissions.txt
ls -l lab6/permissions.txt
```

Q5. Change the permissions of permissions.txt using numeric mode 755 and verify the result.

```
chmod 755 lab6/permissions.txt
ls -l lab6/permissions.txt
```

Q6. Remove execute permission from the owner of permissions.txt using symbolic chmod notation.

```
chmod u-x lab6/permissions.txt
```

Q7. Add read and execute permissions for the group of permissions.txt without changing the owner's permissions.

```
chmod g+rx lab6/permissions.txt
```

Q8. Remove all permissions from others for permissions.txt and verify the result.

```
chmod o-rwx lab6/permissions.txt
ls -l lab6/permissions.txt
```

Q9. Create a directory named securedata and set its permissions to 750 using numeric notation.

```
mkdir securedata
chmod 750 securedata
```

Q10. Explain and demonstrate the difference between file and directory read, write and execute permissions by testing access to a practice directory.

```
# Directory 'r' allows listing contents (ls), 'w' allows creating/deleting files, 'x' allows accessing/entering directory (cd).
# File 'r' allows reading contents, 'w' allows editing, 'x' allows execution as a script.
chmod 755 practice_dir
```

Q11. Display the current umask value and record the result.

```
umask
```

Q12. Temporarily change the umask value to 027 and create a new file and directory. Observe and verify their resulting permissions.

```
umask 027
```

touch testfile.txt

mkdir testdir

ls -ld testfile.txt testdir

Q13. Restore the umask to its previous value and verify the current umask.

umask 022

umask

Q14. Create a new user and group for ACL practice, then create a file named aclfile.txt for testing.

sudo useradd acluser

sudo groupadd aclgroup

touch aclfile.txt

Q15. Display the current ACL of aclfile.txt using getfacl and identify the owner, group, user, group and mask entries.

getfacl aclfile.txt

Q16. Use setfacl to give a specified user read and write permission on aclfile.txt without changing the normal owner/group permission bits. Verify using getfacl.

setfacl -m u:acluser:rw aclfile.txt

getfacl aclfile.txt

Q17. Use setfacl to give a specified group read permission on aclfile.txt and verify the ACL.

setfacl -m g:aclgroup:r aclfile.txt

getfacl aclfile.txt

Q18. Modify and remove a specific ACL entry from aclfile.txt using setfacl. Verify the changes using getfacl.

Modify user permission:

setfacl -m u:acluser:r aclfile.txt

Remove user ACL entry:

setfacl -x u:acluser aclfile.txt

getfacl aclfile.txt

Q19. Set a default ACL on a directory so that newly created files inside it inherit the required access permissions. Create a test file and verify the inherited ACL.

mkdir acldir

setfacl -d -m u:acluser:rw acldir

touch acldir/newfile.txt

getfacl acldir/newfile.txt

Q20. Create, modify, verify and clean up file permissions and ACLs in a complete workflow using chmod, umask, setfacl and getfacl.

mkdir workflow_dir

touch workflow_dir/file.txt

chmod 750 workflow_dir/file.txt

umask 027

```
setfacl -m u:acluser:rwx workflow_dir/file.txt
getfacl workflow_dir/file.txt
setfacl -b workflow_dir/file.txt
rm -rf workflow_dir
```